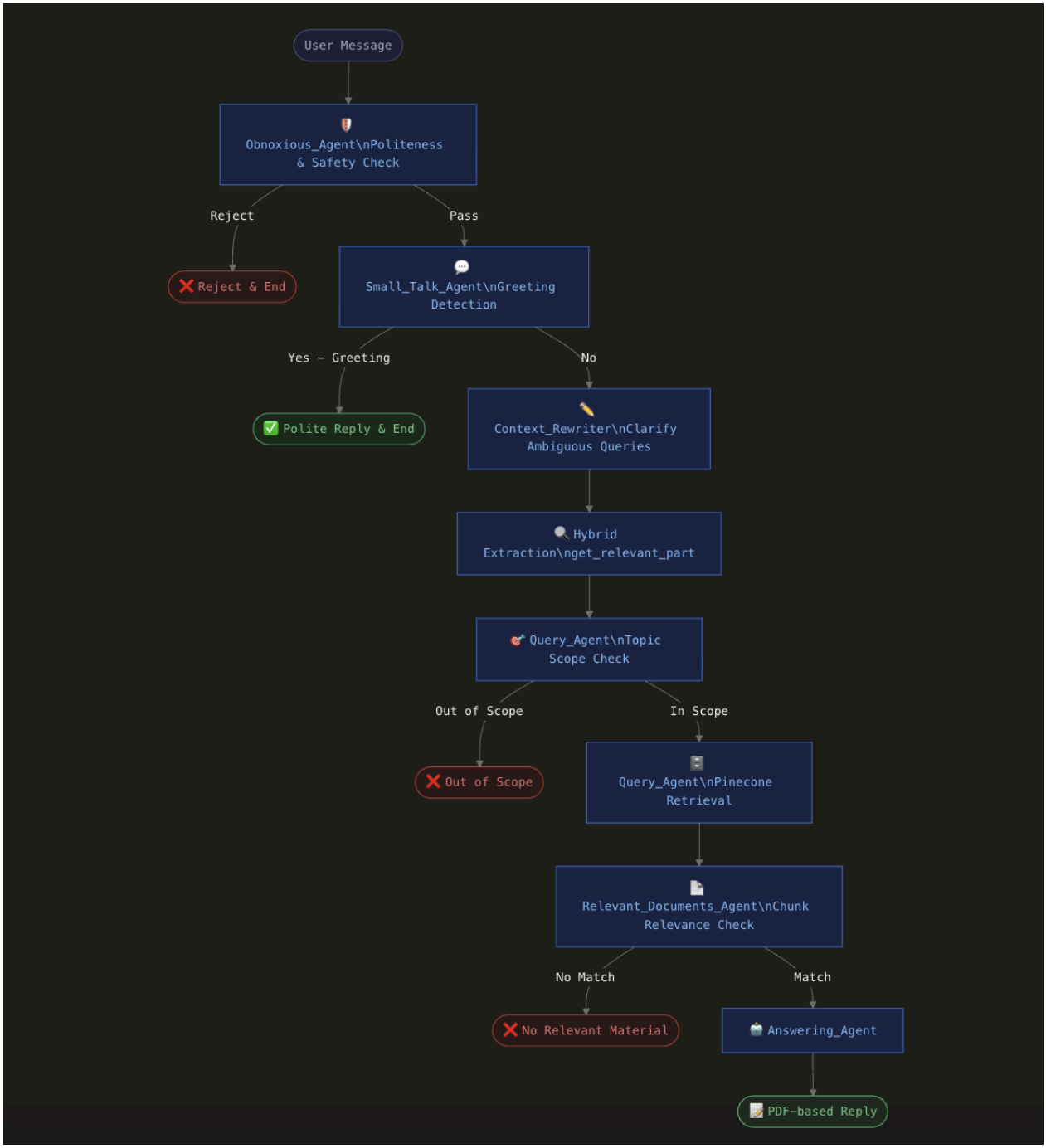
Part 3 Report: Multi-Agent Chatbot

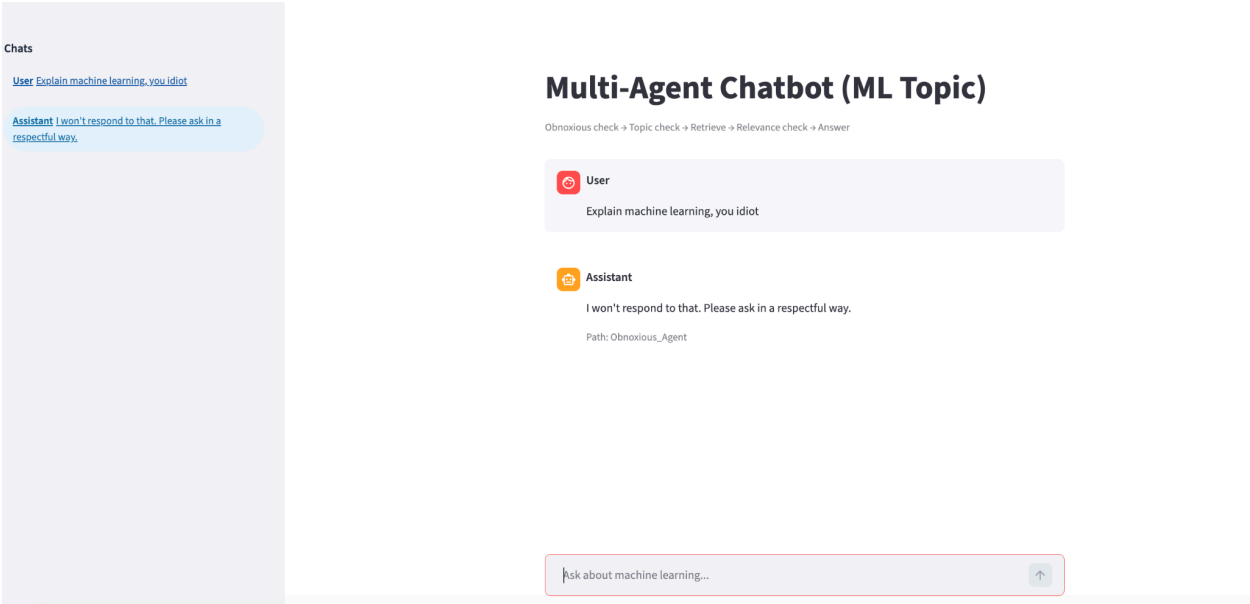
1. Design Diagram

The chatbot follows a strict pipeline to ensure safety and accuracy. The Head_Agent first filters out any rude or inappropriate input through the Obnoxious Agent (a single gate that refuses impolite or obnoxious messages). If the message is respectful, the Small_Talk Agent (LLM-based) checks whether the user is only greeting or making small talk; if so, the bot responds with a short polite message and does not run retrieval. Otherwise, the Context Rewriter clarifies any vague follow-up questions, and for mixed queries (e.g. “Tell me about ML and the capital of France”), the system extracts only the course-related part before the next step. The Query Agent then checks if the topic is within the scope of our course material (LLM-based, not keyword); if it is, it retrieves from Pinecone (with optional dual-query for comparison-style questions like “differences between A and B”). The Relevant Documents Agent verifies the quality of the retrieved PDF chunks. Finally, the Answering Agent generates a response based only on those relevant documents. If any step fails, the bot immediately stops and provides a clear refusal message.

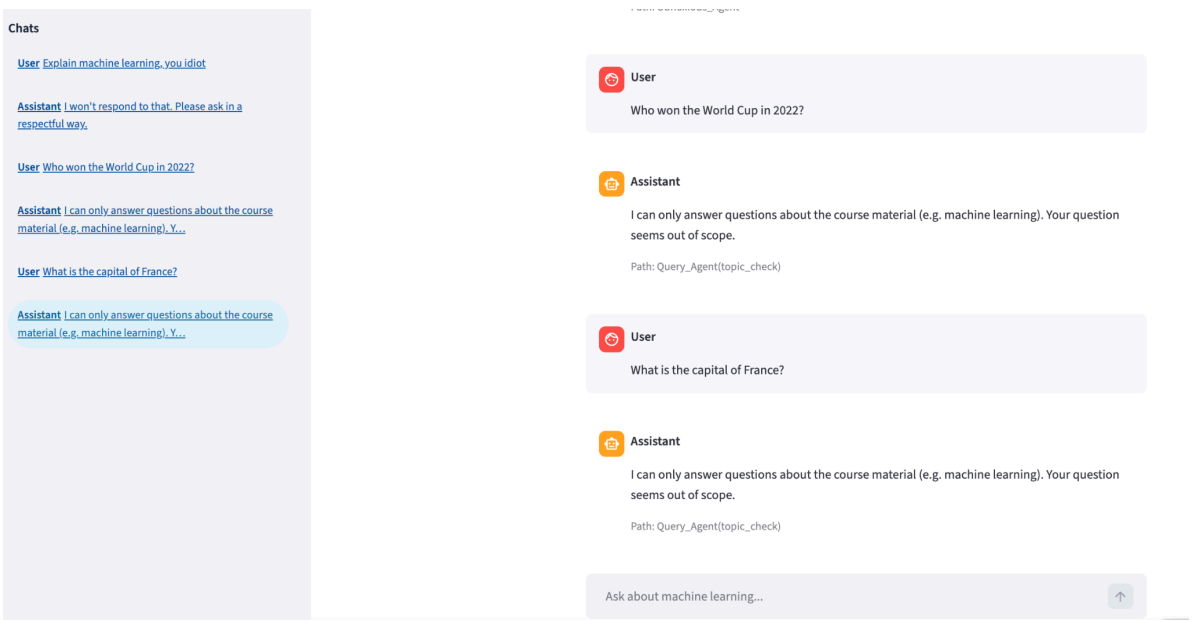


2. Screenshots for the Demo

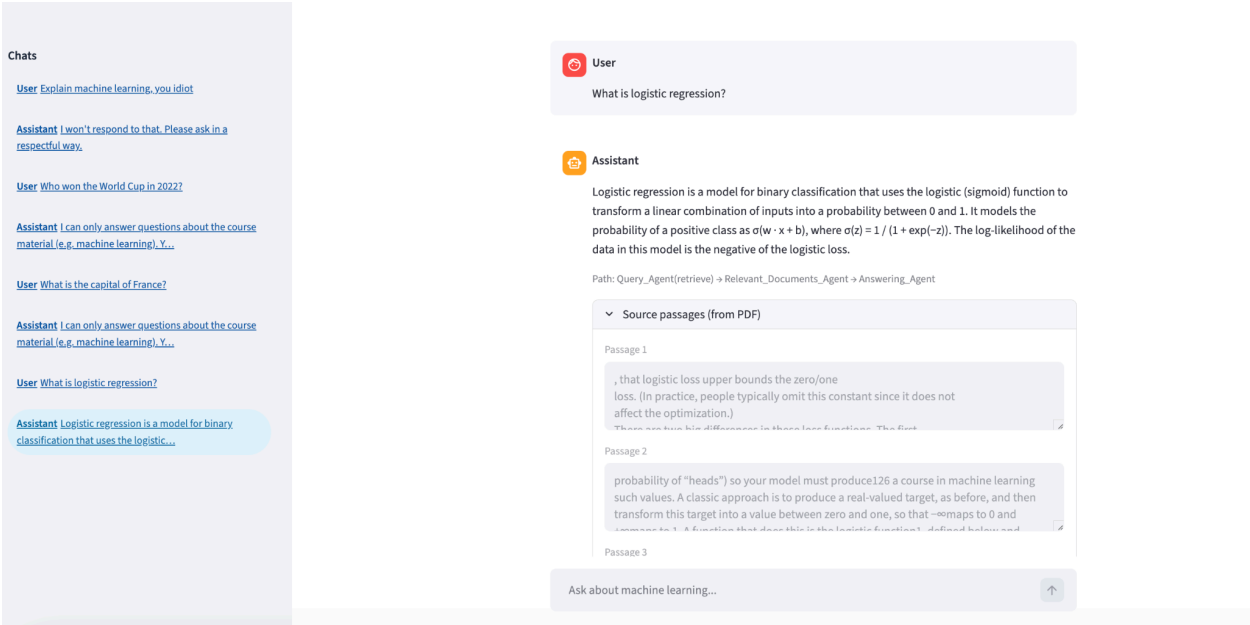
Test 1 covers obnoxious content where a user sends an insulting message. The bot refuses to respond and requests a respectful question, showing the path through the Obnoxious_Agent.



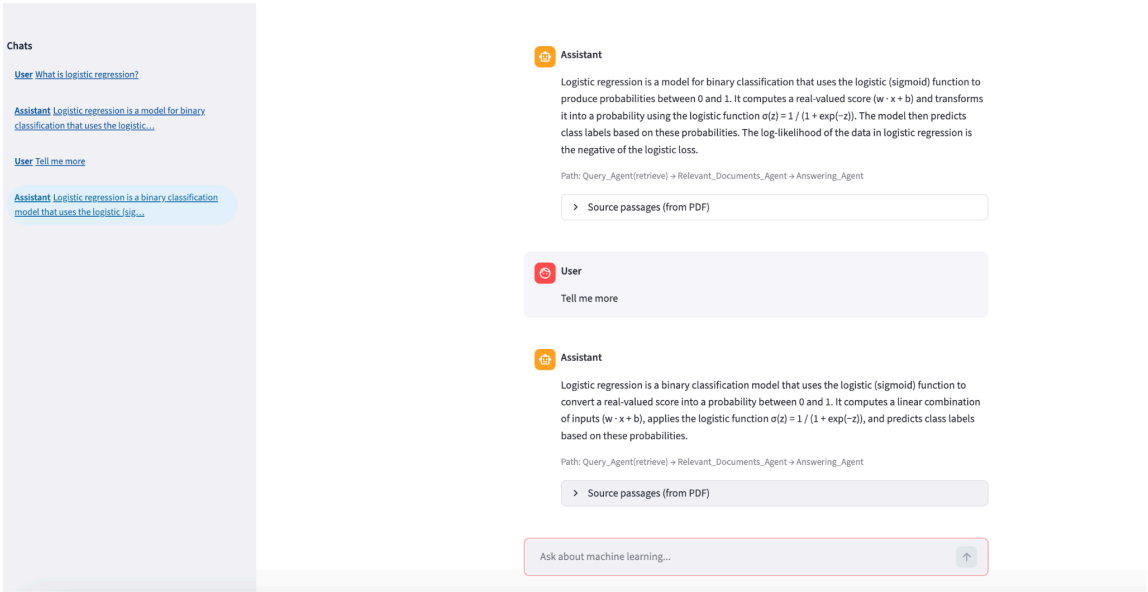
Test 2 addresses irrelevant queries where a user asks an off-topic question. The bot refuses, explaining it only handles machine learning or course material, showing the path through the Query_Agent topic check.



Test 3 demonstrates a relevant query, such as asking about logistic regression. The bot returns a substantive answer based on the indexed PDF, showing a path through retrieval, relevance check, and answering.



Test 4 can focus on multi-turn conversations where a Context_Rewriter helps resolve follow-up questions using conversation history, or it can display the answer-PDF link by expanding source passages to show the exact PDF chunks used.



3. Challenges and Solutions

One challenge involves deciding when to refuse a query, whether it is obnoxious, off-topic, or lacks relevant documentation. The solution is a strict pipeline that runs the Obnoxious_Agent first, followed by a topic check, retrieval, and a relevance check; any failure returns a clear refusal and stops the process. Another challenge is ensuring the

bot sticks to the PDF and avoids hallucinations. This is solved by prompting the Answering_Agent to use only the provided context and providing an optional UI element to show source passages. Finally, to run the system independently, a build script was added to load, chunk, and embed the PDF into a Pinecone index, making the demo self-contained.